

Asset Location Optimizer

- Multi-period optimization model solving optimal asset placement across Taxable, TDA, and Roth accounts.
- Validated across horizons of 1, 5, 10, and 15 years.
- Benchmarked against a naive equal-allocation baseline; alpha computed automatically on every run.
- Parameterized for any horizon, any account balance split, any tax rates, and any contribution and withdrawal schedule.

Tax Realism Built Into the Model

- Capital gains tax on realized gains, with cost basis tracked per asset through every period.
- Ordinary income tax on TDA withdrawals and interest income, using a piecewise marginal bracket model.
- Dividend income modeled separately from price returns.
- Marginal tax rate determined each period by actual income, not assumed fixed.
- Basis updated through contributions, sells, and distributions at every step.

All Ten Factors Addressed

Factor	Implementation
Number and type of accounts	Taxable, TDA, and Roth modeled with distinct tax treatment.
Account balance distributions	Arbitrary initial balances; optimizer relocates freely.
Number of asset classes	Stock, Bond, Cash; basis tracked per asset.
Expected returns	Per-asset, per-period; dividend and interest yields separate.
Tax rates	Capital gains, marginal income brackets, TDA liquidation rate.
Contributions and withdrawals	Per-account, per-period; limits enforced.
Investment horizon	Configurable T; terminal after-tax wealth maximized.
Tax efficiency and unrealized gains	Cost basis tracked; unrealized gain exposure drives placement.
Rebalancing frequency	Annual rebalance; turnover rate controls forced realization.
Bequest considerations	TDA liquidation tax applied at horizon; Roth passes tax-free.

Alpha on a Representative Portfolio (T = 15)

Portfolio and assumptions

Total portfolio	\$720,000 — Taxable 38%, TDA 49%, Roth 13%
Horizon	15 years
Capital gains tax	15%
Annual stock turnover	8%
Naive baseline	60 / 30 / 10 held identically in every account

Result

Optimizer terminal wealth	\$1,869,942
Naive baseline terminal wealth	\$1,845,983
Alpha	\$23,959 (1.30% over 15 years, or \$1,597 per year)

Optimizer placed stocks predominantly in TDA, with bonds and cash in Roth — reflecting standard location theory, confirmed by the model.

PIP Deliverables Met

- Working optimizer; runs end-to-end without manual intervention.
- All ten PIP outcome factors implemented and demonstrable.
- Naive benchmark comparison built in; alpha computed on every run.
- Validated at $T = 1, 5, 10$, and 15 years.
- Live runner allows scenario changes without modifying code.
- Code is documented and reproducible.